



JKBOSE 2026

CLASS 11TH

CONIC SECTIONS

MATHS

LECTURE-1

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TOPICS TO BE COVERED

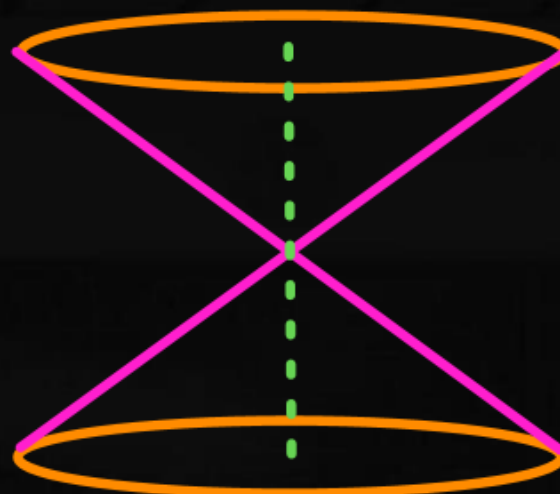
1 What is a Conic Section?

2 Circle

3 Related Problems

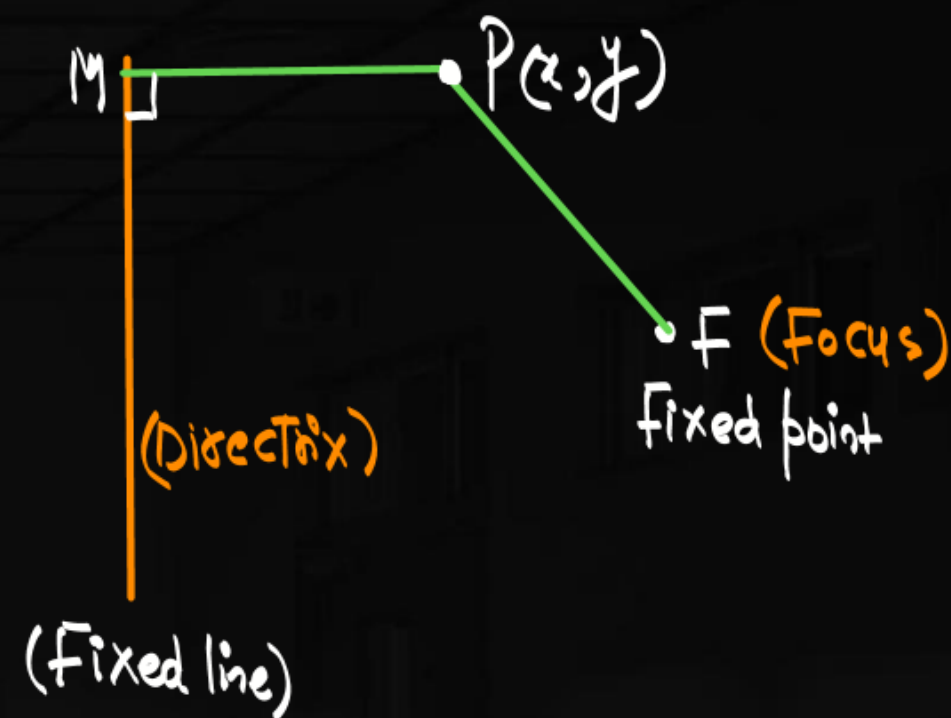
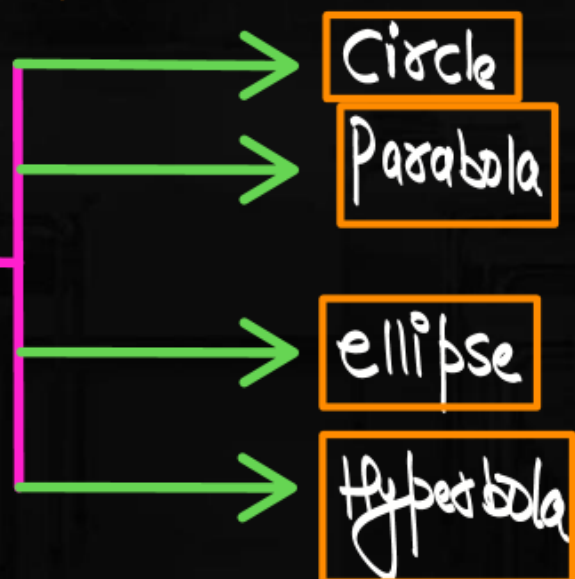
INTRODUCTION

Conic section: Section of the Double napped Cone by the plane
i.e. the Curves obtained by intersecting a plane with the Double napped



Cone (Right circular cone) are known as Conic Sections or Conics.

Conic sections



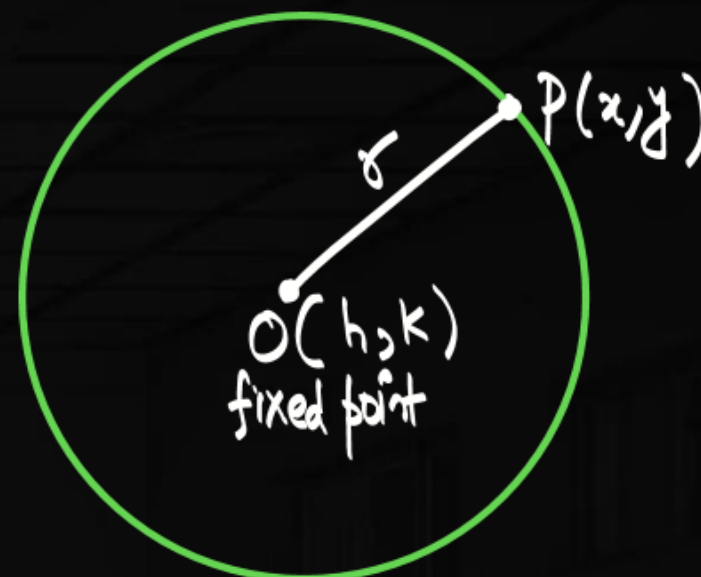
* The locus of a point in a plane such that the ratio of its distance from a fixed point to its perpendicular distance from the fixed straight line is constant, is known as the Conic Section.
i.e. $\frac{PF}{PM} = \text{Constant} = e$.

- * The fixed point is known as the focus of the conic section.
- * The fixed line is known as the Directrix of the conic section.
- * The Constant ratio is known as the eccentricity of conic section.

Based on the values of e , we have the following:

- * $e = 1 \longrightarrow$ Parabola
- * $0 < e < 1 \longrightarrow$ ellipse
- * $e > 1 \longrightarrow$ Hyperbola

Circle:



The locus of a point in a plane such that its distance from fixed point is always constant, is known as a circle.
 or The collection of points in plane which are equidistant from a fixed point is known as a circle.

- * Fixed point is known as the Centre of the circle.
 - * Constant distance is known as the radius of the circle.
- Suppose that $P(x, y)$ be any point on the circle with $O(h, k)$ & radius r . So $OP = r$.
- $$\Rightarrow OP^2 = r^2$$

$$\Rightarrow \sqrt{(x-h)^2 + (y-k)^2} = r$$

$$\Rightarrow (x-h)^2 + (y-k)^2 = r^2 \quad \text{Which is the required equation of the circle.}$$

Example: Find the equation of the circle with centre $(3, -2)$ and radius 4 units.

(Sol) Required equation of the circle is:

$$(x-h)^2 + (y-k)^2 = r^2$$

$$\Rightarrow (x-3)^2 + (y+2)^2 = 4^2$$

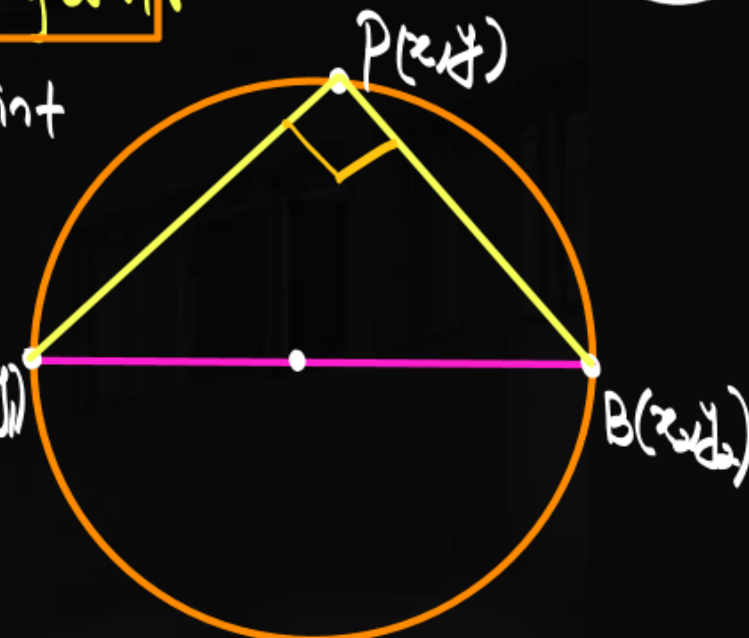
$$\Rightarrow x^2 + 9 - 6x + y^2 + 4 + 4y = 16$$

$$\Rightarrow x^2 + y^2 - 6x + 4y - 3 = 0$$



Equation of the Circle in Diameter form:

Suppose that $P(x, y)$ be any point on the circle with AB as its diameter, where $A(x_1, y_1)$ & $B(x_2, y_2)$ are the ends of the diameter.



Here $\angle APB = 90^\circ$ (Angle in a semi-circle)

$$\Rightarrow AP \perp PB, \text{ so } m_{AP} \times m_{PB} = -1$$

$$\Rightarrow \left(\frac{y-y_1}{x-x_1} \right) \left(\frac{y-y_2}{x-x_2} \right) = -1$$

$$\Rightarrow \frac{(y-y_1)(y-y_2)}{(x-x_1)(x-x_2)} = -1$$

$$\Rightarrow (y-y_1)(y-y_2) = -(x-x_1)(x-x_2)$$

$$\Rightarrow (x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$$

Example: Find the equation of the circle whose ends of diameters are $A(3, -2)$ & $B(4, 5)$.

(Sol) Required equation is:

$$(x-3)(x-4) + (y+2)(y-5) = 0.$$

$$\Rightarrow x^2 - 7x + 12 + y^2 - 3y - 10 = 0$$

$$\Rightarrow x^2 + y^2 - 7x - 3y + 2 = 0$$

Hence the result.

EXAMPLE

Find the equation of circle having center at $(2, 3)$ and touching the Line $x + y = 1$.

(Sol) Here $OP = r$ r represents the distance of the point $O(2, 3)$ from the line $x + y - 1 = 0$.

$$\text{So, } OP = \frac{|2+3-1|}{\sqrt{1^2+1^2}} = \frac{4}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{4\sqrt{2}}{2}$$

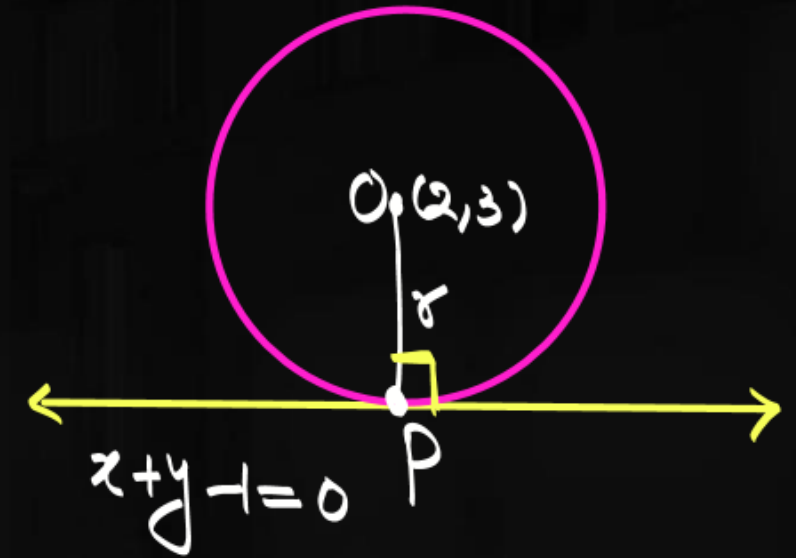
$$\Rightarrow r = 2\sqrt{2}$$

Therefore the equation of the Circle is:

$$(x-2)^2 + (y-3)^2 = (2\sqrt{2})^2$$

$$\Rightarrow x^2 + 4 - 4x + y^2 + 9 - 6y = 8$$

$$\Rightarrow x^2 + y^2 - 4x - 6y + 5 = 0$$



QUESTION (NCERT EXAMPLER):

The area of the circle centred at $(1, 2)$ and passing through $(4, 6)$ is.

1 5π ✗

2 10π ✗

3 25π ✓✓

4 None of these

(Sol) Suppose that the equation of the required circle be:

$$(x-1)^2 + (y-2)^2 = r^2 \text{ --- (1)}$$

Since $(4, 6)$ lies on (1), so

$$(4-1)^2 + (6-2)^2 = r^2$$

$$\Rightarrow r^2 = 9 + 16$$

$$\Rightarrow r^2 = 25$$

$$\Rightarrow r = 5$$

So Required Area = $\pi r^2 = 25\pi$ Sq. units.



General Form of the equation of a Circle

Consider the following equation:

$$x^2 + y^2 + 2gx + 2fy + c = 0 \quad \text{--- (1)}$$

Adding on b/s by $g^2 + f^2$ on (1), we get;

$$x^2 + y^2 + 2gx + 2fy + c + g^2 + f^2 = g^2 + f^2$$

$$\Rightarrow (x^2 + 2gx + g^2) + (y^2 + 2fy + f^2) = g^2 + f^2 - c$$

$$\Rightarrow (x+g)^2 + (y+f)^2 = (\sqrt{g^2 + f^2 - c})^2 \quad (g^2 + f^2 - c > 0)$$

Comparing it with $(x-h)^2 + (y-k)^2 = r^2$.

$$\Rightarrow -h = g, -k = f, r = \sqrt{g^2 + f^2 - c}$$

$$\Rightarrow h = -g, k = -f, r = \sqrt{g^2 + f^2 - c}$$

This shows that (1) represents a circle in the general form with centre $(-g, -f) = (-\frac{\text{coeff of } x}{2}, -\frac{\text{coeff of } y}{2})$ & radius $r = \sqrt{g^2 + f^2 - c}$.

Example: Find the radius and centre of the circle $x^2 + y^2 + 10x + 4y - 1 = 0$.

Sol) Here $2g = 10, 2f = 4, c = -1$.

$$\Rightarrow g = 5, f = 2, c = -1$$

$$\text{So, centre} = (-g, -f) = (-5, -2)$$

$$\& \text{ Radius} = \sqrt{25 + 4 + 1} = \sqrt{30}$$

Hence the result.

EXAMPLE

Find the centre and the radius of the circle $x^2 + y^2 + 8x + 10y - 8 = 0$.

Homework
Problem.

EXAMPLE



If $x^2 + y^2 - 2x + 2ay + a + 3 = 0$ represents the real circle with non Zero radius then find the values of a .

Sol) Given equation is:

$$x^2 + y^2 - 2x + 2ay + a + 3 = 0 \quad \text{--- (1)}$$

Here radius, $r = \sqrt{1 + a^2 - a - 3}$

$$\Rightarrow r = \sqrt{a^2 - a - 2}$$

For (1) to represent a real circle with non-zero radius,

$$a^2 - a - 2 > 0$$

$$\Rightarrow (a+1)(a-2) > 0$$

$$\Rightarrow a < -1 \text{ or } a > 2$$

$$\begin{array}{c} \oplus \quad \ominus \quad \oplus \\ \hline -1 \quad 2 \end{array}$$

$$\Rightarrow a \in (-\infty, -1) \cup (2, \infty)$$

Therefore the possible values of a are such that $a < -1$ or $a > 2$.

Hence the result.

QUESTION (NCERT EXAMPLER):

The equation of the circle having centre $(1, -2)$ and passing through the point of intersection of the lines $3x + y = 14$ and $2x + 5y = 18$ is.

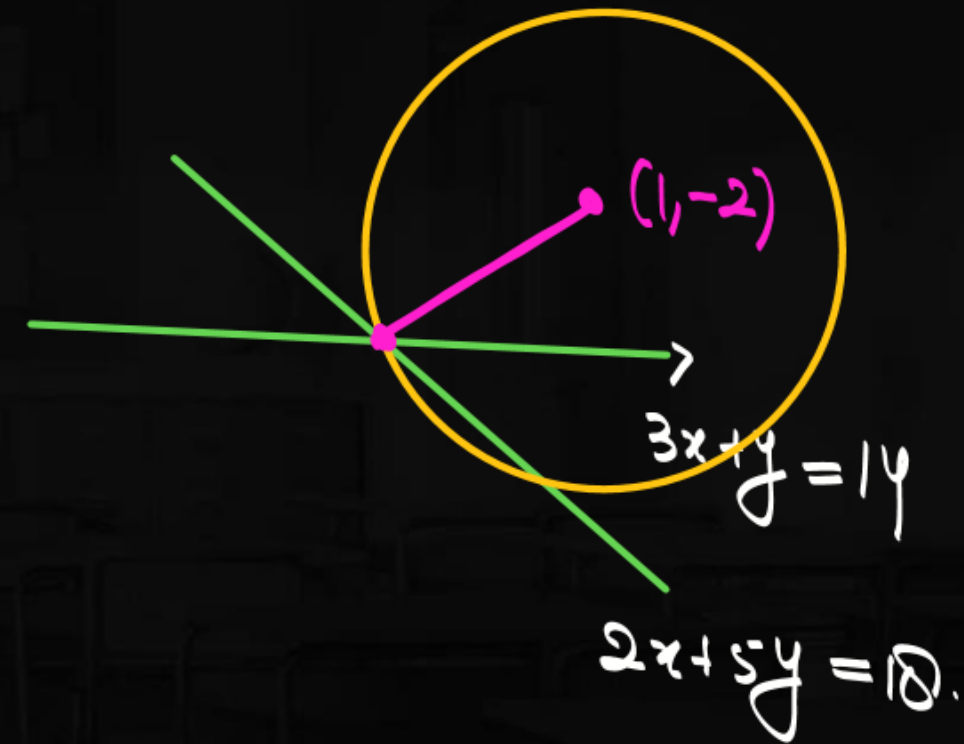
(Homework Problem)

1 $x^2 + y^2 - 2x + 4y - 20 = 0$

2 $x^2 + y^2 - 2x - 4y - 20 = 0$

3 $x^2 + y^2 + 2x - 4y - 20 = 0$

4 $x^2 + y^2 + 2x + 4y - 20 = 0$



Taukeer Amin Sofi 

Huzair Maqbool 

Rasib Ayoub Rather 

Thank

You